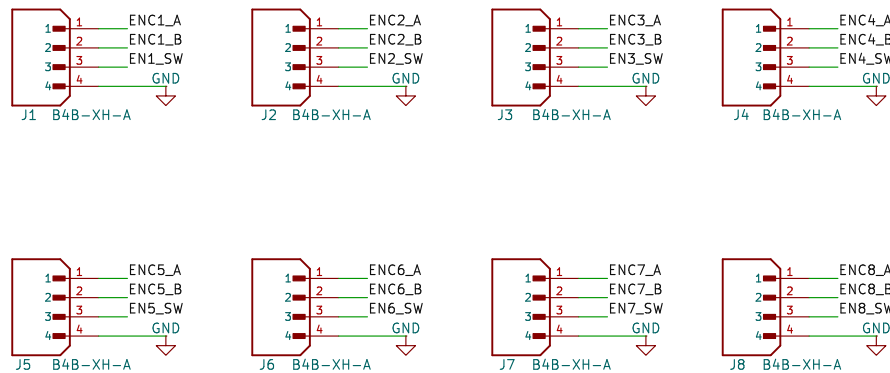
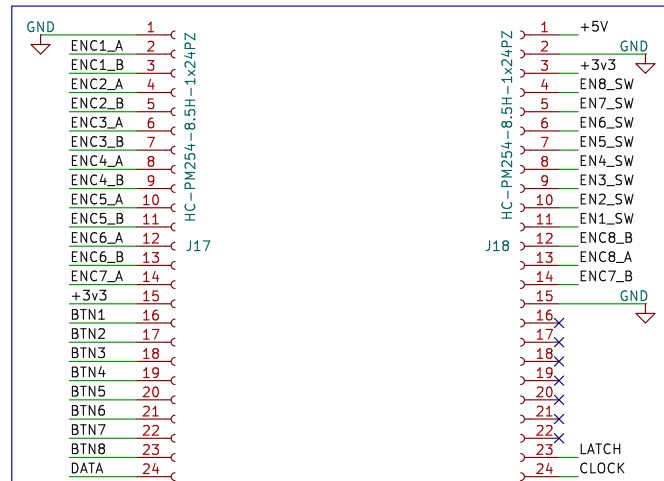
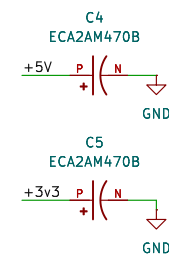


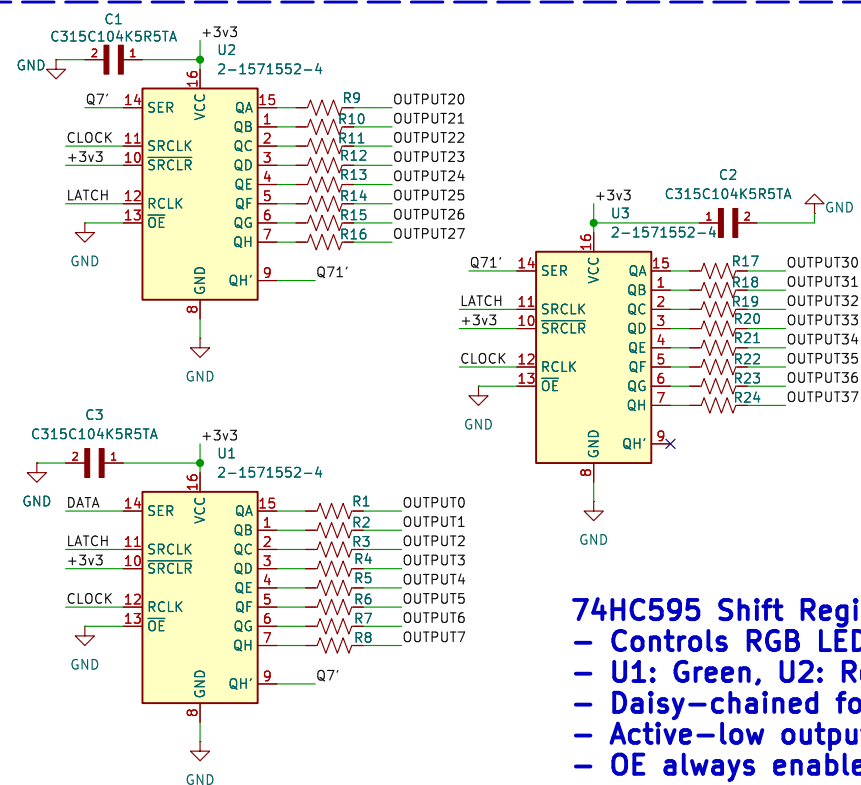


- Rotary Encoders:**
- Each has Channel A/B + push switch
  - Signals to Teensy GPIOs with internal pull-ups
  - Detects rotation direction/speed
  - Push switch provides extra input

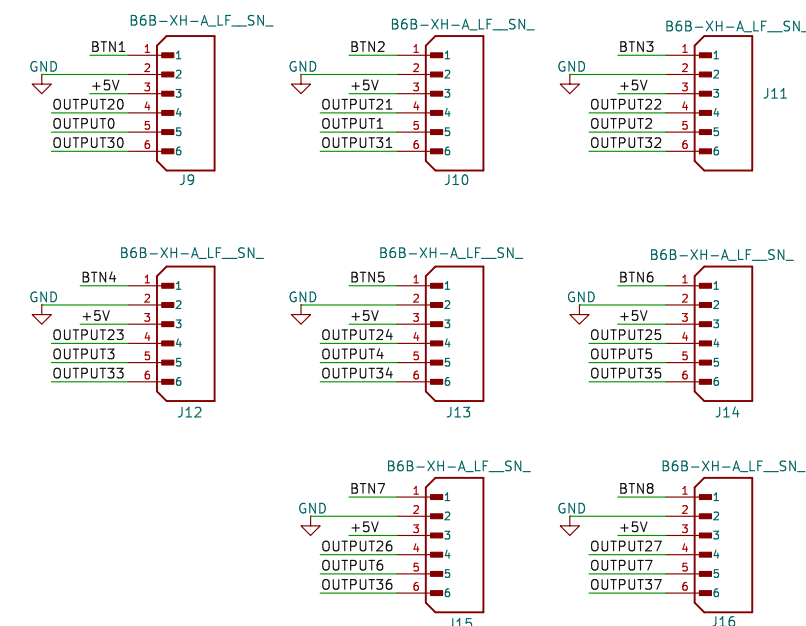


Teensy 4.1\_MCU\_48PINS\_SOCKET

- Teensy 4.1 – Microcontroller:**
- Main controller for MIDI system
  - Reads 8 rotary encoders (A/B + push) and 8 RGB buttons
  - Drives RGB LEDs via three 74HC595 shift registers (DATA/CLOCK/LATCH)
  - Provides +3.3V logic, +5V LED power
  - Inputs active-low with internal pull-ups



- 74HC595 Shift Registers – RGB LED Driver:**
- Controls RGB LED cathodes
  - U1: Green, U2: Red, U3: Blue
  - Daisy-chained for 24 outputs via 3 pins
  - Active-low outputs (LOW = LED ON)
  - OE always enabled, MR tied HIGH



- RGB Buttons:**
- Each has 1 tactile switch + 3 LEDs (R/G/B)
  - Switches active-low to Teensy
  - LED anodes to +5V, cathodes via 220Ω to shift registers
  - Provides user input and LED feedback